Supported Targets	ESP32	ESP32-C2	ESP32-C3	ESP32-C5	ESP32-C6	ESP32-C61	ESP32-H2	ESP32-P4	ESP32-S2	ESP32-S3	Linux

Hello World Example

Starts a FreeRTOS task to print "Hello World".

(See the README.md file in the upper level 'examples' directory for more information about examples.)

How to use example

Follow detailed instructions provided specifically for this example.

Select the instructions depending on Espressif chip installed on your development board:

- ESP32 Getting Started Guide
- ESP32-S2 Getting Started Guide

Example folder contents

The project **hello_world** contains one source file in C language hello_world_main.c. The file is located in folder main.

ESP-IDF projects are built using CMake. The project build configuration is contained in `CMakeLists.txt` files that provide set of directives and instructions describing the project's source files and targets (executable, library, or both).

Below is short explanation of remaining files in the project folder.

```
├── CMakeLists.txt
├── pytest_hello_world.py      Python script used for automated testing
├── main
│   ├── CMakeLists.txt
│   └── hello_world_main.c
└── README.md                  This is the file you are currently reading
```

For more information on structure and contents of ESP-IDF projects, please refer to Section Build System of the ESP-IDF Programming Guide.

Troubleshooting

- Program upload failure
  - Hardware connection is not correct: run `idf.py -p PORT monitor`, and reboot your board to see if there are any output logs.
  - The baud rate for downloading is too high: lower your baud rate in the `menuconfig` menu, and try again.

Technical support and feedback

Please use the following feedback channels:

- For technical queries, go to the esp32.com forum
- For a feature request or bug report, create a GitHub issue

We will get back to you as soon as possible.

Result

```
linux x86 is little endian!
sizeof(null): 8
sizeof(char): 1
sizeof(short): 2
sizeof(int): 4
sizeof(long int): 4
sizeof(long long int): 8
sizeof(float): 4
sizeof(double): 8
sizeof(char *): 8
sizeof(size_t): 8
sizeof(ssize_t): 8
sizeof(data): 11
sizeof(data_p): 8

var a is greater than b! char is SIGNED!!!

data allocated at: 00007FF7D5CB4010
data points to: 00007FF7D5CB4010
data contents: "3DUfw·
data[0]:  as hex: 0x11
data[1]: " as hex: 0x22
data[2]: 3 as hex: 0x33
data[3]: D as hex: 0x44
data[4]: U as hex: 0x55
data[5]: f as hex: 0x66
data[6]: w as hex: 0x77
data[7]: ·as hex: 0xffffff88
data[8]: ·as hex: 0xffffff99
data[9]:  as hex: 0x00

data_as_int, allocated at: 000000762C9FF7D0
data as int points to: 00007FF7D5CB4010
data_as_int[0]: 44332211
data_as_int[1]: 88776655
data_as_int[2]: 99

location: 00007FF7D5CB4011, value: 55443322
float location: 00007FF7D5CB4011, value: 13482743300096.000000
```